Workshop 3: Renewable Energy Design Challenge

Reflection

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Design Development

The first five basic types of appliances were used in the house, and I started with a total daily demand of 13.10 kWh and a peak of 2.06 kW. I then used 16 types of appliances in my household, including two air conditioners, one water heater, one washing machine, one dishwasher, entertainment devices, and computing loads. This increase in appliance variety raised the peak demand to greater than 5.6 kW and the daily consumption to approximately 33-40 kWh, better reflecting a realistic four-occupant household.

The solar array was originally designed with 10 panels, which produced much more energy than required but did not address the mismatch between energy production during the day and consumption during the evening and night hours. Adding more panels to the configuration had only a small effect on coverage (from 58% to approximately 71%) while pushing the system cost close to the budget limit, indicating that panel count was no longer the limiting factor - battery capacity was.

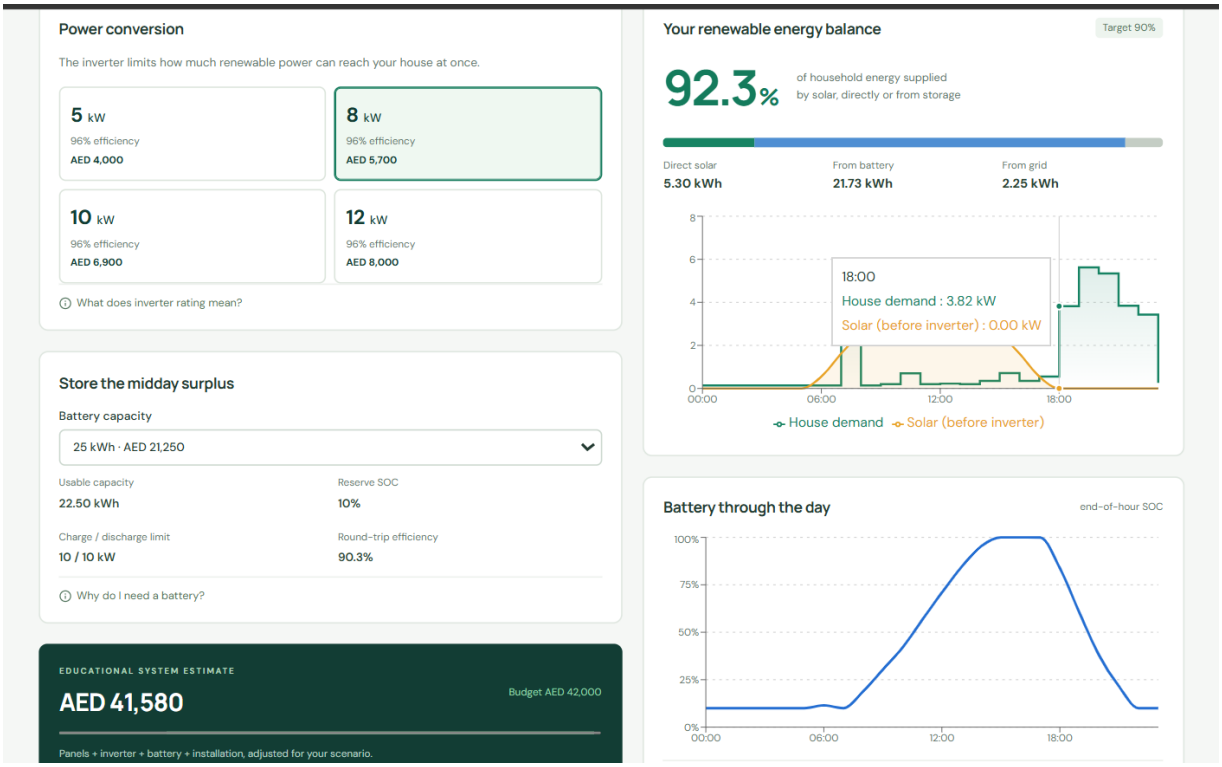
I then reduced the panel count to a more efficient level and focused on sizing the battery and inverter. I chose an 8 kW inverter to comfortably cover the peak load of 5.62 kW without overloading, since a 5 kW inverter would have been insufficient. I then tested battery sizes from 5 kWh up to 25 kWh. Every increment in battery size resulted in a significant increase in renewable coverage, which confirmed that the binding constraint for this household's demand profile was storage, not generation. Lastly, I reduced the wattage and run-hours of some appliances slightly to keep total demand lower while maintaining a realistic level of usage. This brought the design to 92.3% renewable coverage at a system cost of AED 41,580, within the AED 42,000 budget limit.

Evaluation

The final design was successful in achieving both engineering objectives: 92.3% renewable coverage (above the 90% target) and a total system cost of AED 41,580 (within the AED 42,000 budget). The major trade-off I identified was between the size of the solar array and the size of the battery. Beyond a certain point, extra panels created excess energy that could not be consumed during peak demand periods of the day (midday) and instead had to be exported or wasted, since demand actually peaks in the evening (from 18:00 onward) due to air

conditioning, lighting, and water heating. Battery capacity was the most expensive component per kWh, but once the array was scaled to an appropriate size, additional battery capacity provided a much larger increase in coverage per dirham spent than additional panels did.

With a bigger budget, I would prioritize increasing battery size (30+ kWh) over adding more panels, since storage is what truly limits this system. Another option worth exploring in future design iterations would be to shift flexible loads, such as the washing machine or water heater, to daytime hours when solar power is most available, as this would reduce reliance on the battery during the evening and could allow a smaller, less expensive battery to achieve the same coverage target.



Harvest the sun.

Turn available roof space into a solar array. Balance generation, space and cost.

YOUR MISSION
90% renewable
4 occupants · AED 42,000 budget

Peak load	Daily consumption	PV capacity	Battery capacity	Renewable coverage	System estimate
5.62 kW	29.27 kWh	6.75 kWp	25.00 kWh	92.3 %	41,580 AED

Choose your solar module

400 W / MODULE	450 W / MODULE	550 W / MODULE
Standard 400	**Balanced 450**	**High-output 550**
1.75 × 1.1 m	1.9 × 1.13 m	2.28 × 1.13 m
AED 500	AED 610	AED 760

Panel orientation: Portrait

Number of panels: 15

0 panels — 35 panels physically fit

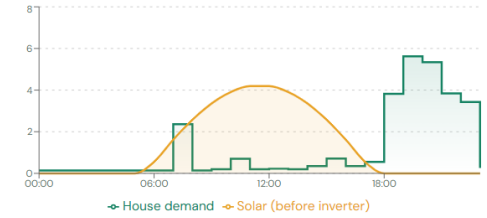
Your roof, from above

34.4% utilized

← 10 m →



Sunlight meets demand



House energy 29.27 kWh

Solar generation 32.40 kWh

What are peak sun hours?

YOUR SOLAR RESOURCE

Al Ain

6

Lighting · 0.36 kWh/day

Quantity: 6 | Rated power · W: 10 | 6.00 h/day — Edit schedule

Refrigerator
Refrigeration · 1.44 kWh/day

Quantity: 1 | Rated power · W: 150 | 9.60 h/day — Edit schedule

Laptop
Computing · 0.30 kWh/day

Quantity: 1 | Rated power · W: 60 | 5.00 h/day — Edit schedule

Air conditioner
Cooling · 15.00 kWh/day

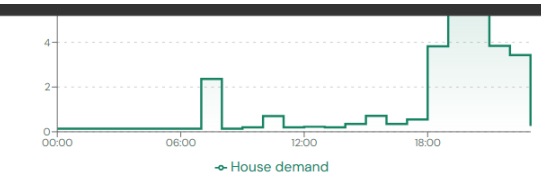
Quantity: 2 | Rated power · W: 1500 | 5.00 h/day — Edit schedule

Water heater
Water heating · 3.60 kWh/day

Quantity: 1 | Rated power · W: 1800 | 2.00 h/day — Edit schedule

Ceiling light group
Lighting · 0.36 kWh/day

Quantity: 6 | Rated power · W: 10 | 6.00 h/day — Edit schedule



Daily energy **29.27 kWh**

Peak demand **5.62 kW**

Power or energy?

Where does the energy go?

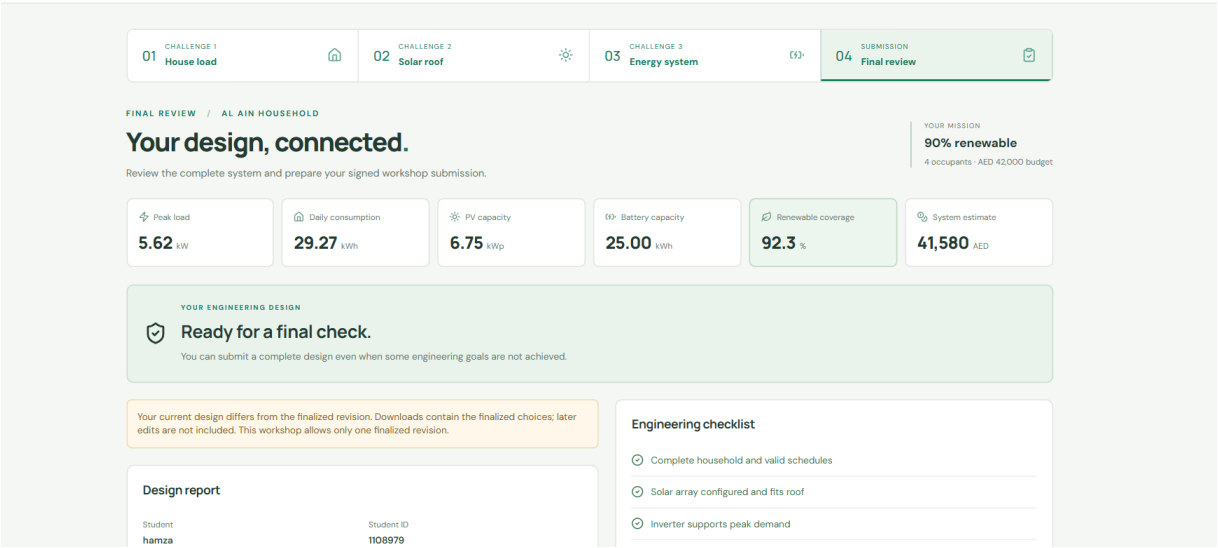


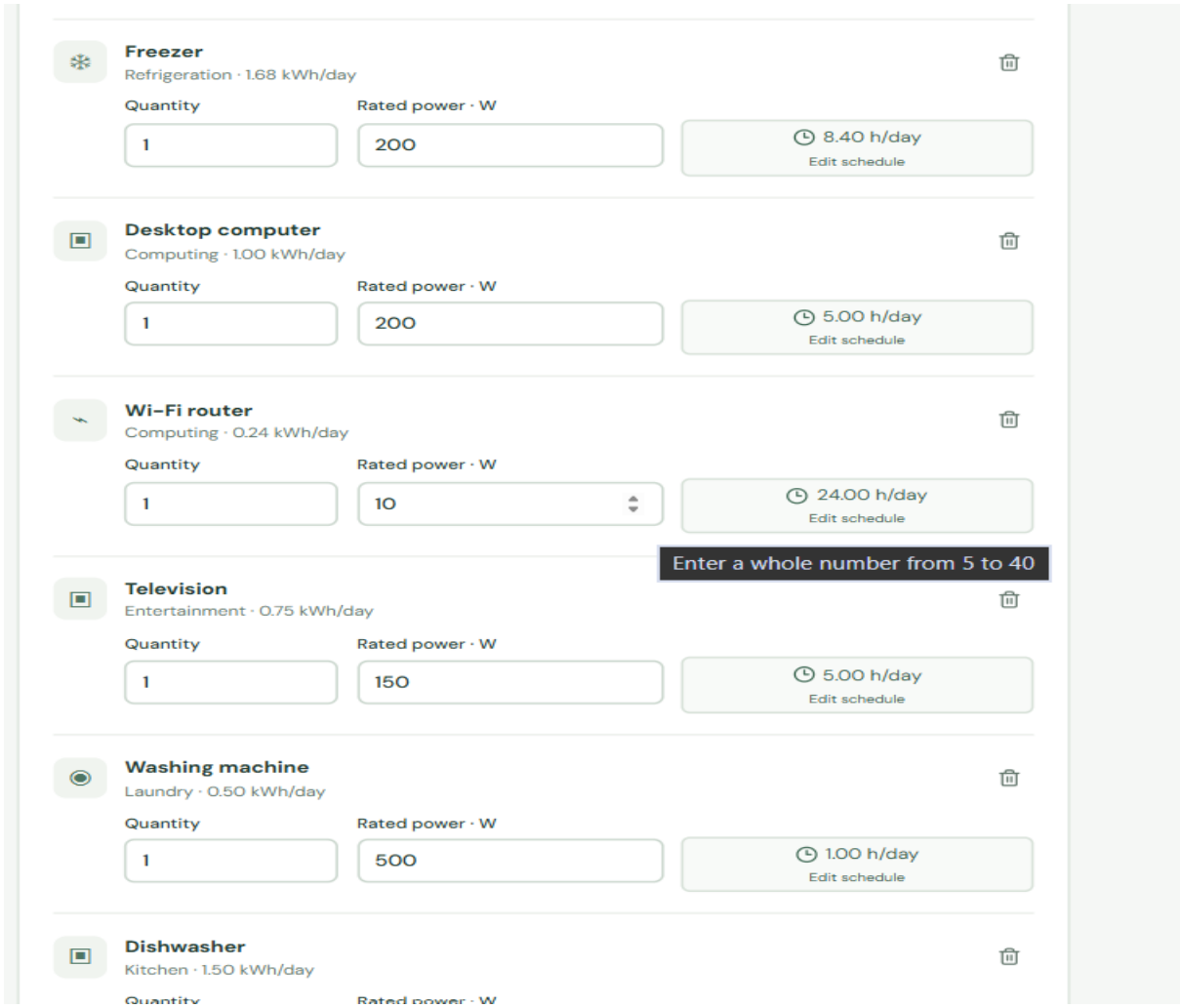
Lighting 2.5% | Refrigeration 10.7% | Computing 5.5% | Cooling 57.0% | Water heating 12.3% | Entertainment 2.6% | Laundry 1.7% | Kitchen 5.1% | Cooking 2.7%

ENGINEERING NOTE 01

Timing changes your peak.

Two appliances can use the same daily energy but place very different demands on the system when they run together.


01 CHALLENGE 1 House load
02 CHALLENGE 2 Solar roof
03 CHALLENGE 3 Energy system
04 SUBMISSION Final review
FINAL REVIEW / AL AIN HOUSEHOLD
Your design, connected.
Review the complete system and prepare your signed workshop submission.
YOUR MISSION
90% renewable
4 occupants · AED 42,000 budget
Peak load 5.62 kW
Daily consumption 29.27 kWh
PV capacity 6.75 kWp
Battery capacity 25.00 kWh
Renewable coverage 92.3 %
System estimate 41,580 AED
YOUR ENGINEERING DESIGN
Ready for a final check.
You can submit a complete design even when some engineering goals are not achieved.
Your current design differs from the finalized revision. Downloads contain the finalized choices; later edits are not included. This workshop allows only one finalized revision.
Engineering checklist
Complete household and valid schedules
Solar array configured and fits roof
Inverter supports peak demand
Design report
Student
hamza
Student ID
1108979


Freezer
Refrigeration · 1.68 kWh/day
Quantity
1
Rated power · W
200
8.40 h/day
Edit schedule
Desktop computer
Computing · 1.00 kWh/day
Quantity
1
Rated power · W
200
5.00 h/day
Edit schedule
Wi-Fi router
Computing · 0.24 kWh/day
Quantity
1
Rated power · W
10
24.00 h/day
Edit schedule
Enter a whole number from 5 to 40
Television
Entertainment · 0.75 kWh/day
Quantity
1
Rated power · W
150
5.00 h/day
Edit schedule
Washing machine
Laundry · 0.50 kWh/day
Quantity
1
Rated power · W
500
1.00 h/day
Edit schedule
Dishwasher
Kitchen · 1.50 kWh/day

ube Google خرائط ترجمة Discord — Downlo... Redeem Code Amazon.co.uk – O... Booking.com

Dishwasher
Kitchen · 1.50 kWh/day

Quantity	Rated power · W	
1	1500	1.00 h/day Edit schedule

Electric kettle
Cooking · 0.42 kWh/day

Quantity	Rated power · W	
1	1400	0.30 h/day Edit schedule

Microwave
Cooking · 0.38 kWh/day

Quantity	Rated power · W	
1	1200	0.32 h/day Edit schedule

Ceiling fan
Cooling · 1.68 kWh/day

Quantity	Rated power · W	
3	70	8.00 h/day Edit schedule

Phone charger
Computing · 0.06 kWh/day

Quantity	Rated power · W	
4	5	3.00 h/day Edit schedule

Choose an appliance... + Add